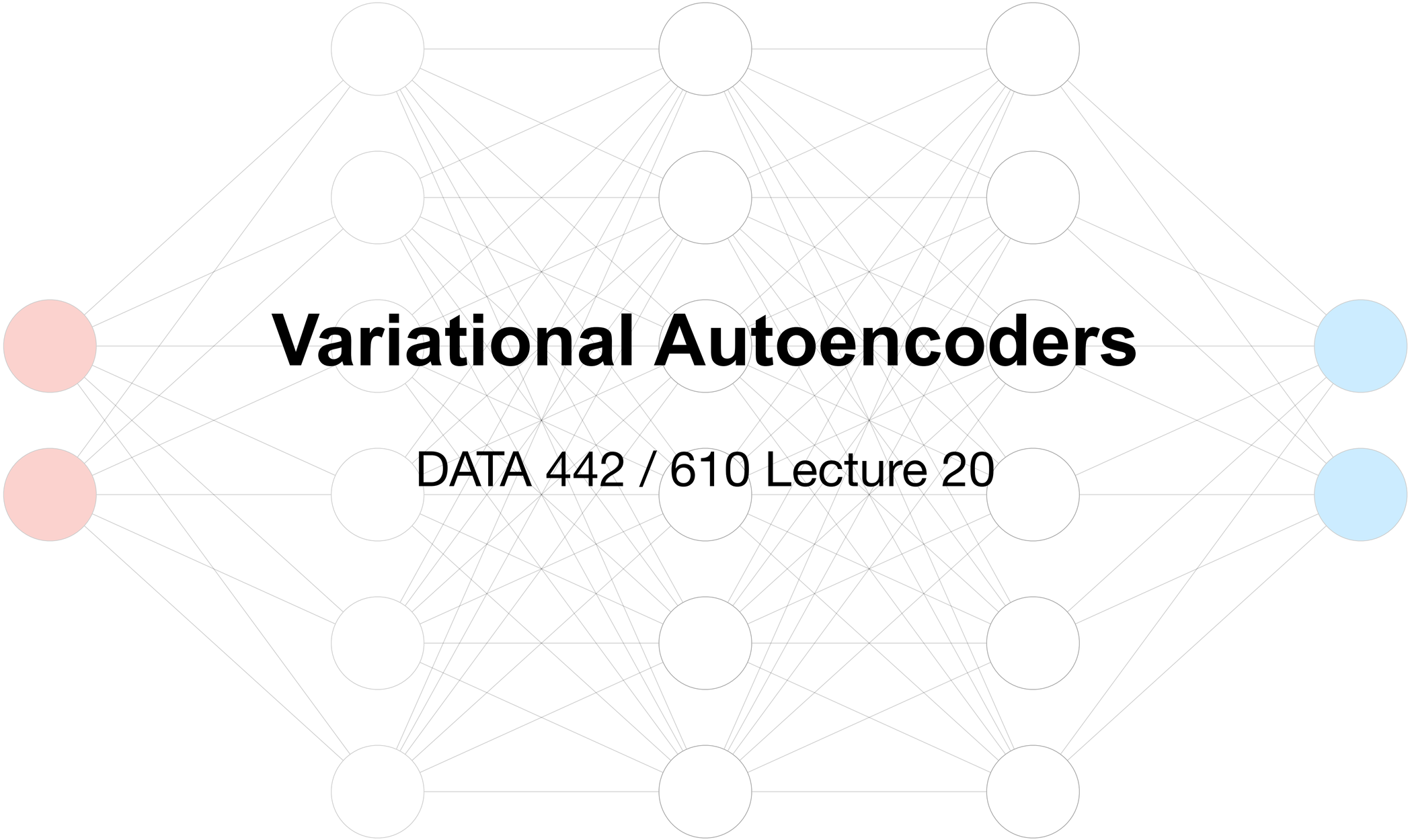


Variational Autoencoders

DATA 442 / 610 Lecture 20

Outline

- Review of representations
- Variational autoencoders
  - Evidence lower bound (ELBO)
  - Reparameterization trick
  - Calculating KL Divergence
- What next?

Relevant Reading: *Deep Learning* Chapter 15, Alex Alemi's notes, ArXiv.1312.6114

How do we learn representations?

1. Draw a diagram of the data we have P
2. Augment the diagram with representations
3. Draw a diagram of how we would like to model the data and representations Q
4. Minimize the **Kullback-Leibler Divergence** $D_{KL}(P \parallel Q)$
5. Adjust seasonings, salt to taste

Kullback-Leibler Divergence

Definition: $D_{KL}\left[P \parallel Q\right] \equiv \mathbb{E}_{x\sim p(x)}\left[\log \frac{p(x)}{q(x)}\right] \geq 0$

What is it?

A measure of distance between two distributions p, q

A measure of “surprise” when observing samples drawn from p if expecting a different distribution q

A measure of our ability to discern the “real world” p from a proposed model q

Example: Supervised learning with representations

$$D_{KL}\big(P \,\|\, Q\big) = D_{KL}\big(p(x,y)\; p_\theta(z\,|\,x) \;\|\, p(x)\, q(z)\, q_\phi(y\,|\,z)\big)$$

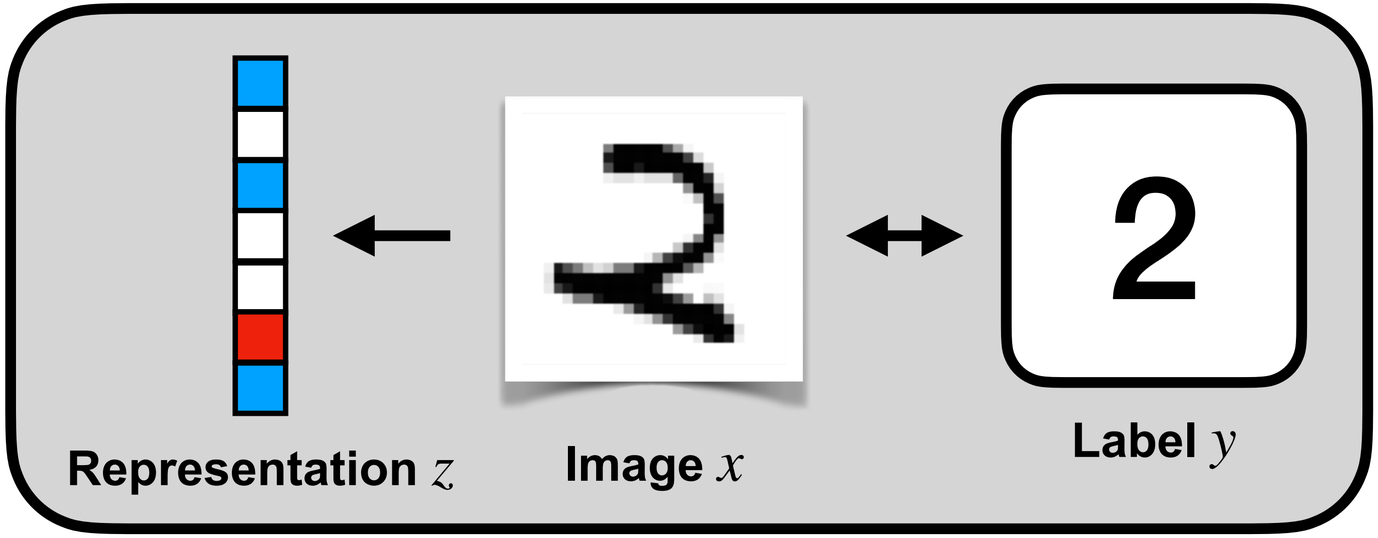
$$= \mathbb{E}_{x,y\sim p; z\sim p_\theta}\left[\log \frac{p(x,y)\, p_\theta(z\,|\,x)}{p(x)\, q(z)\, q_\phi(y\,|\,z)}\right]$$

$$= \mathbb{E}_{x,y\sim p; z\sim p_\theta}\left[\log \frac{p(y\,|\,x)\, p_\theta(z\,|\,x)}{q(z)\, q_\phi(y\,|\,z)}\right]$$

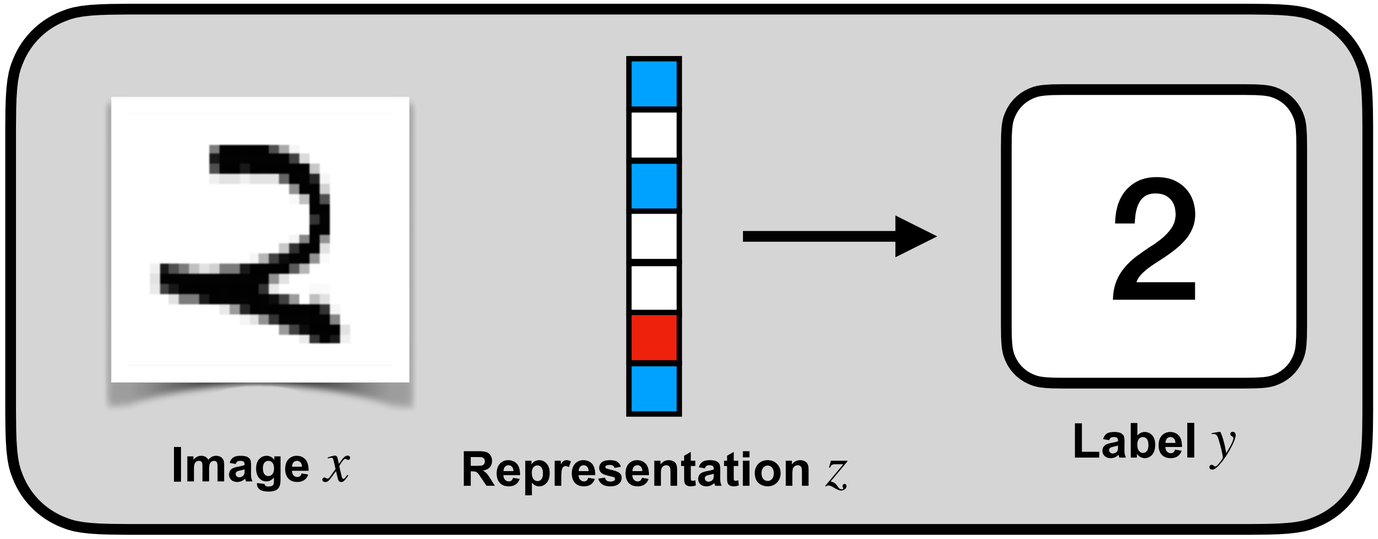
$$= \mathbb{E}_{x,y\sim p; z\sim p_\theta}\big[\log p(y\,|\,x)\big]$$
$$- \mathbb{E}_{x,y\sim p; z\sim p_\theta}\big[\log q_\phi(y\,|\,z)\big]$$
$$+ \mathbb{E}_{x,y\sim p; z\sim p_\theta}\left[\log \frac{p_\theta(z\,|\,x)}{q(z)}\right]$$

$$\mathscr{L} = - \mathbb{E}_{x,y\sim p; z\sim p_\theta}\big[\log q_\phi(y\,|\,z)\big] + D_{KL}\big[p_\theta(z\,|\,x) \,\|\, q(z)\big]$$

Real world P : $p(x,y,z) = p(x,y)\, p_\theta(z\,|\,x)$



Fantasy world: Q : $q(x,y,z) = p(x)\, q(z)\, q_\phi(y\,|\,z)$



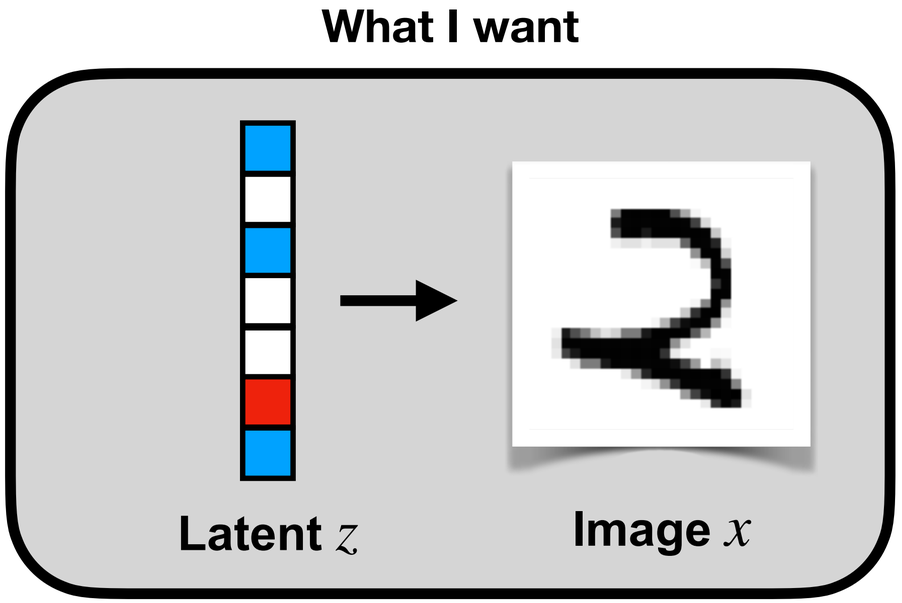
Latent generative models

Goal: data generative model $p_\theta(x)$

Strategy: try to model generation process with latent variable $z \sim p_\theta(z),\ z \rightarrow x$

$$p_\theta(x) = \int dz\ p_\theta(x, z)$$

$$= \int dz\ p_\theta(x \mid z)\, p_\theta(z)$$



Applications

1. Data generation $x \sim p_\theta(x)$
2. Latent inference $z \sim p_\theta(z \mid x)$

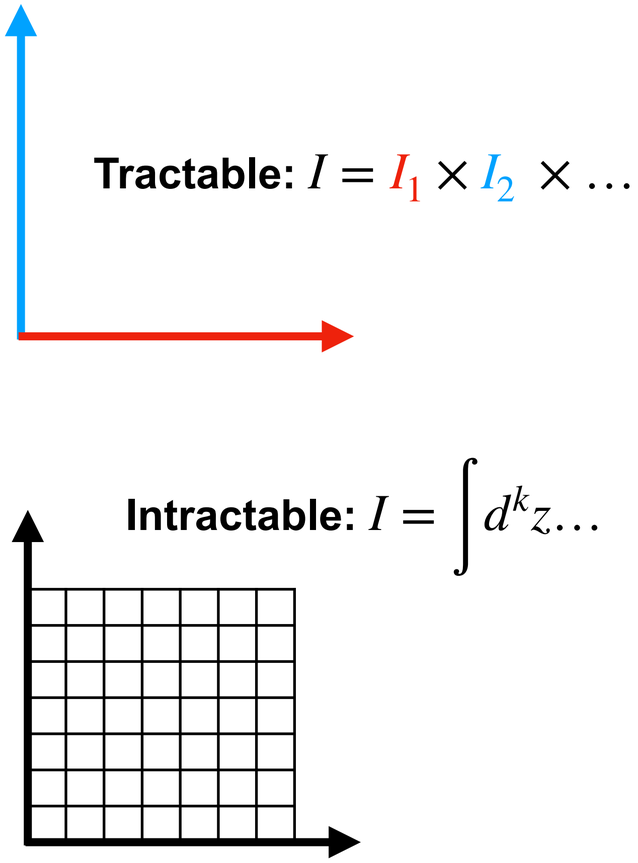
$$p_\theta(z \mid x) = \frac{p_\theta(x \mid z) p_\theta(z)}{p_\theta(x)}$$

Problem: Intractable integrals

Likelihood $p_\theta(x) = \int d\vec{z}\; p_\theta(x \mid \vec{z})\, p_\theta(\vec{z})$ is *really* inconvenient to compute for nonlinear models with continuous latent *vectors* $\vec{z}$

Why? The integral does not factorize cleanly

$$\int d\vec{z}\; p_\theta(x \mid \vec{z})\, p_\theta(\vec{z}) \neq \left(\int dz_1\, f[z_1] \right) \ldots \left(\int dz_k\, f[z_k] \right)$$



The universal recipe for learning representations

1. Draw a diagram of the data we have P
2. Augment the diagram with representations
3. Draw a diagram of how we would like to model the data and representations Q
4. Minimize the **Kullback-Leibler Divergence** $D_{KL}(P \parallel Q)$
5. Adjust seasonings, salt to taste

Example: Variational autoencoders

$$D_{KL}\big(P \,\|\, Q\big) = D_{KL}\big(p(x)\; p_\theta(z\,|\,x)\; \|\, q(z)\, q_\phi(x\,|\,z)\big)$$

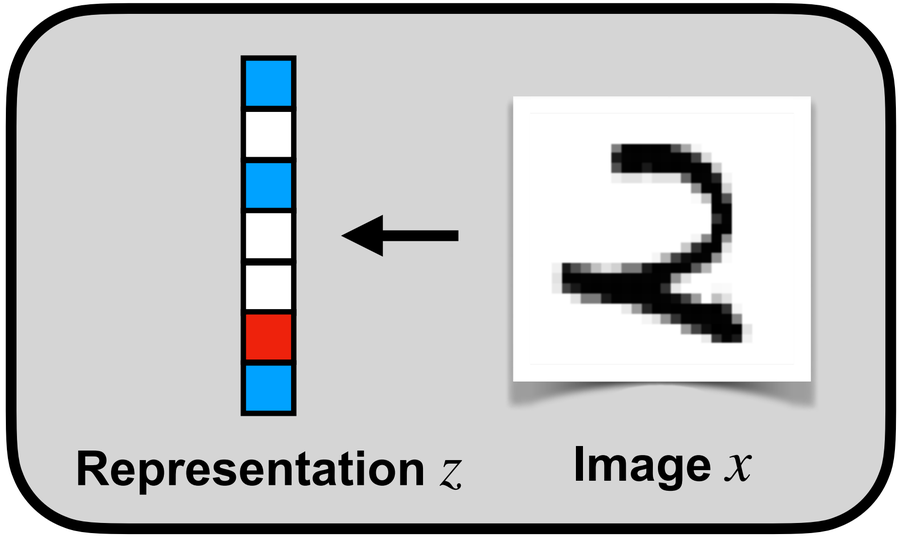
$$= \mathbb{E}_{x\sim p; z\sim p_\theta}\left[\log \frac{p(x)\, p_\theta(z\,|\,x)}{q(z)\, q_\phi(x\,|\,z)}\right]$$

$$= \cancel{\mathbb{E}_{x\sim p; z\sim p_\theta}\big[\log p(x)\big]} - \mathbb{E}_{x\sim p; z\sim p_\theta}\big[\log q_\phi(x\,|\,z)\big] + \mathbb{E}_{x\sim p; z\sim p_\theta} \log \frac{p_\theta(z\,|\,x)}{q(z)}$$

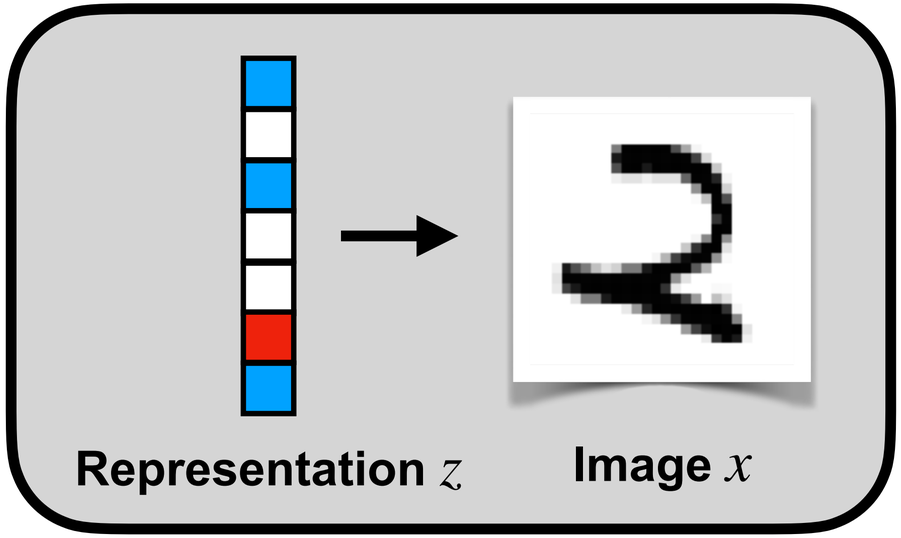
$$\mathscr{L} = -\,\mathbb{E}_{x\sim p; z\sim p_\theta}\big[\log q_\phi(x\,|\,z)\big] + D_{KL}\big[p_\theta(z\,|\,x) \,\|\, q(z)\big]$$

reconstruction **regularization**

Real world P : $p(x, z) = p(x)\, p_\theta(z\,|\,x)$



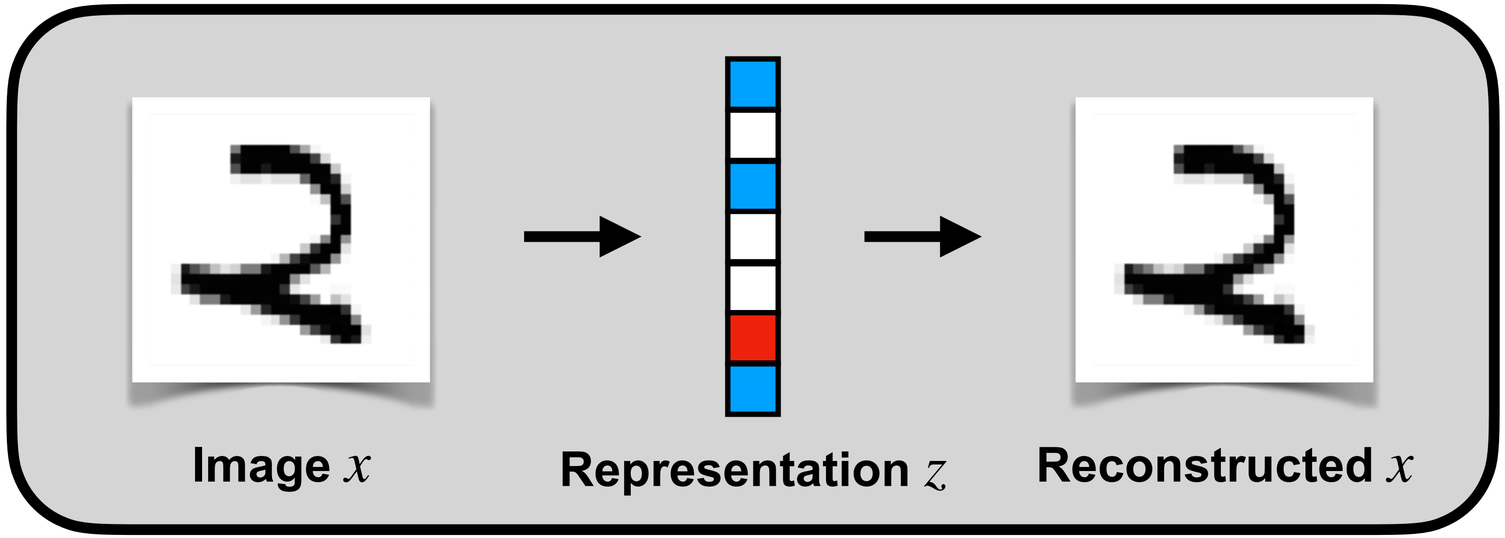
Fantasy world Q : $q(x, z) = q(z)\, q_\phi(x\,|\,z)$



Variational autoencoders

Probabilistic encoder $p_\theta(z \mid x)$

Probabilistic decoder $q_\phi(x \mid z)$



$$\mathscr{L} = -\mathbb{E}_{x\sim p; z\sim p_\theta}\left[\log q_\phi(x \mid z)\right] + D_{KL}\left[p_\theta(z \mid x) \,\|\, q(z)\right]$$

reconstruction **regularization**

Evidence lower bound

Loss: $\mathscr{L} = -\mathbb{E}_{x\sim p; z\sim p_\theta}\left[\log q_\phi(x\,|\,z)\right] + D_{KL}\left[p_\theta(z\,|\,x)\,\|\,q(z)\right]$

$$= -\mathbb{E}\left[\log q_\phi(x\,|\,z)\right] + \mathbb{E}\left[\log p_\theta(z\,|\,x)\right] - \mathbb{E}\left[q(z)\right]$$

$$= -\mathbb{E}\left[\log q_\phi(x, z)\right] + \mathbb{E}\left[\log p_\theta(z\,|\,x)\right]$$

$$= -\mathbb{E}\left[\log q_\phi(z\,|\,x) q_\phi(x)\right] + \mathbb{E}\left[\log p_\theta(z\,|\,x)\right]$$

$$= -\left(\mathbb{E}\left[\log q_\phi(x)\right] - D_{KL}\left[\log p_\theta(z\,|\,x)\,\|\,\log q_\phi(z\,|\,x)\right]\right)$$

Evidence lower bound (ELBO)

$$D_{KL}(\,\cdot\|\cdot\,) \geq 0 \rightarrow \text{ELBO} \leq \mathbb{E}[\log q_\phi(x)]$$

Evidence lower bound

Loss: $\mathscr{L} = -\mathbb{E}_{x\sim p; z\sim p_\theta}\left[\log q_\phi(x\,|\,z)\right] + D_{KL}\left[p_\theta(z\,|\,x)\,\|\,q(z)\right]$

$$= -\mathbb{E}\left[\log q_\phi(x\,|\,z)\right] + \mathbb{E}\left[\log p_\theta(z\,|\,x)\right] - \mathbb{E}\left[q(z)\right]$$

Minimizing the VAE loss *maximizes* a lower bound on the intractable log-evidence

$$= -\left(\mathbb{E}\left[\log q_\phi(x)\right] - D_{KL}\left[\log p_\theta(z\,|\,x)\,\|\,\log q_\phi(z\,|\,x)\right]\right)$$

Evidence lower bound (ELBO)

$D_{KL}(\cdot\|\cdot) \geq 0 \rightarrow$ ELBO $\leq \mathbb{E}[\log q_\phi(x)]$

Okay, but how do I train it?

Loss: $\mathcal{L} = -\mathbb{E}_{x\sim p; z\sim p_\theta}\left[\log q_\phi(x\,|\,z)\right] + D_{KL}\left[p_\theta(z\,|\,x)\,\|\,q(z)\right]$

NLL loss on reconstructed samples

???

$$D_{KL}\left[p_\theta(z\,|\,x)\,\|\,q(z)\right] = \mathbb{E}_{\substack{x\sim p(x);\\ z\sim p_\theta(z|x)}}\left[\log p_\theta(z\,|\,x) - \log q(z)\right]$$

Estimating gradients for this is very brittle

Reparameterization trick: introduce auxiliary random variable ε (not trainable)

$$z \sim p_\theta(z\,|\,x) = \mathcal{N}(\mu_\theta, \sigma_\theta^2)$$
$$= \mu_\theta + \sigma_\theta\,\varepsilon; \quad \varepsilon \sim \mathcal{N}(0,1)$$

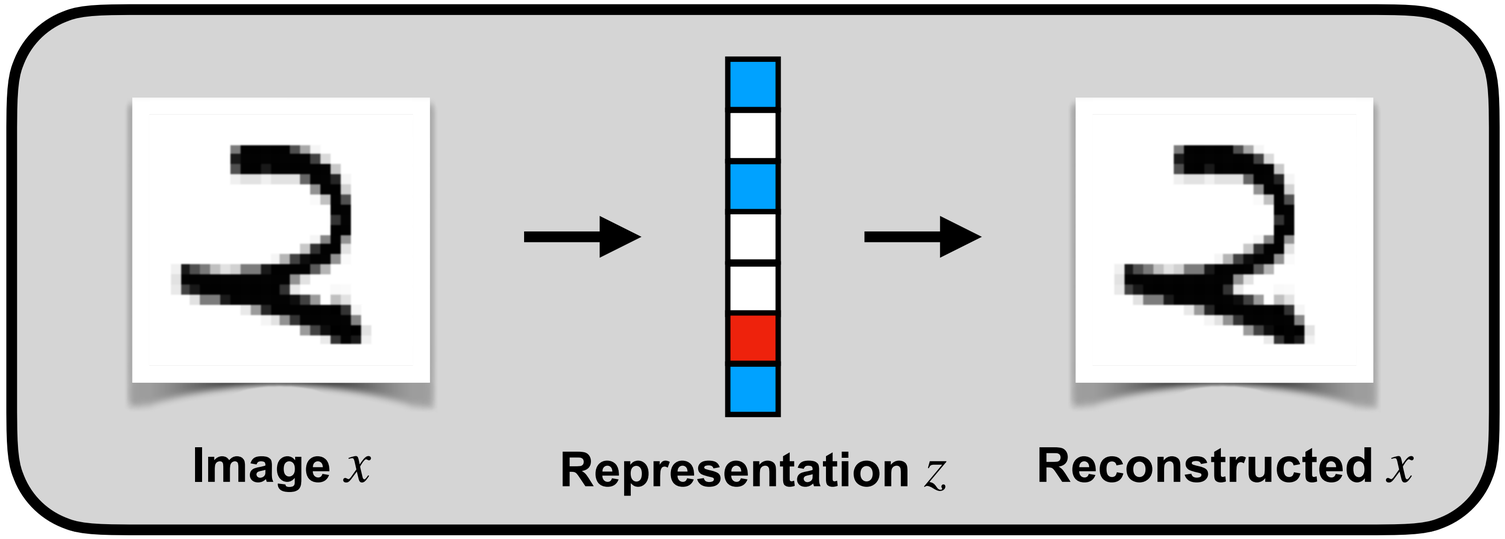
$$\mathbb{E}_{\substack{x\sim p(x);\\ z\sim p_\theta(z|x)}}\left[f(z)\right] \Rightarrow \mathbb{E}_{\substack{x\sim p(x);\\ \varepsilon\sim p(\varepsilon)}}\left[f(x,\varepsilon)\right]$$

Adjust seasonings, salt to taste

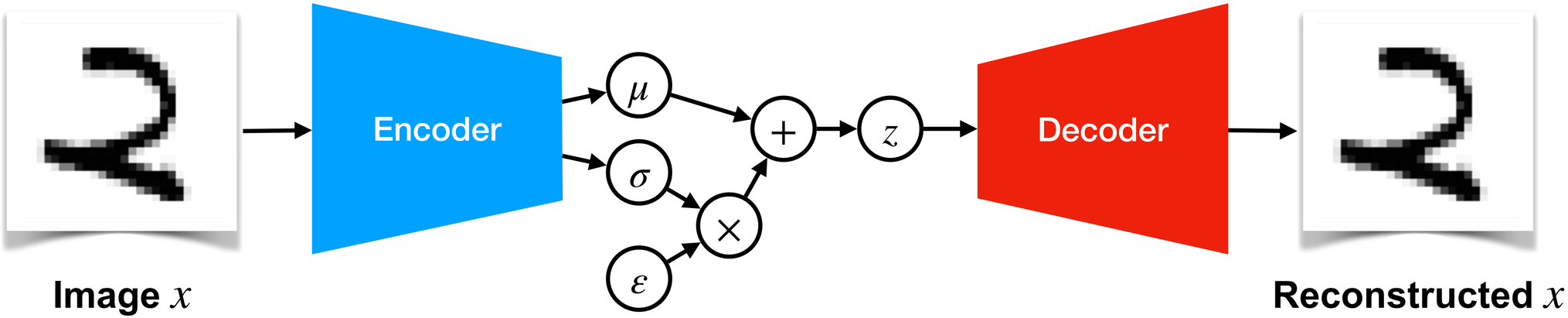
$$\mathcal{L} = -\mathbb{E}_{x\sim p; z\sim p_\theta}\left[\log q_\phi(x\,|\,z)\right] + D_{KL}\left[p_\theta(z\,|\,x)\,\|\,q(z)\right]$$

- KL divergence is a regularization on the learned representations (analogous to L2 regularization via a Gaussian prior on model weights)
- Typical to use a weight $\beta \neq 1$ on the KL Divergence term to balance trade off between prediction and regularization
- **Reparameterization trick:** "sample" $p_\theta(z\,|\,x)$ using auxiliary noise ε
- **Gaussian Prior:** $D_{KL}\left[\mathcal{N}(\mu, \sigma^2)\,\|\,\mathcal{N}(0,1)\right]$ on the board

Variational autoencoders



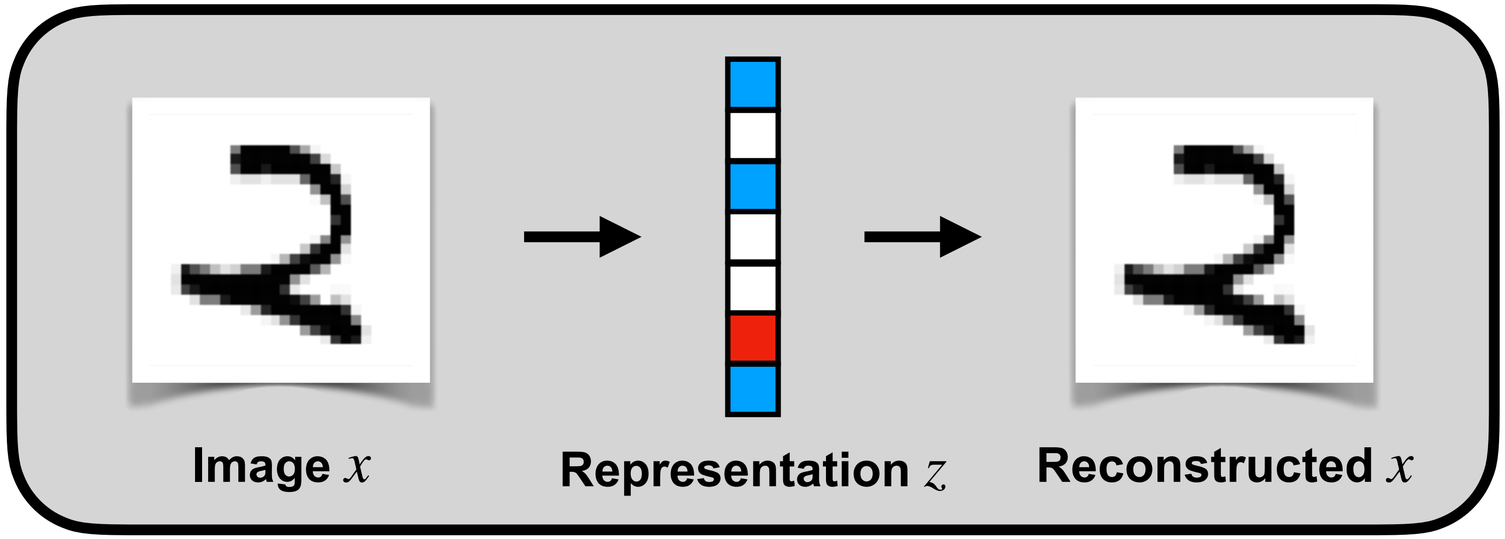
$$\mathscr{L} = -\mathbb{E}_{x\sim p; z\sim p_\theta}\left[\log q_\phi(x\,|\,z)\right] + D_{KL}\left[p_\theta(z\,|\,x)\,\|\,q(z)\right]$$



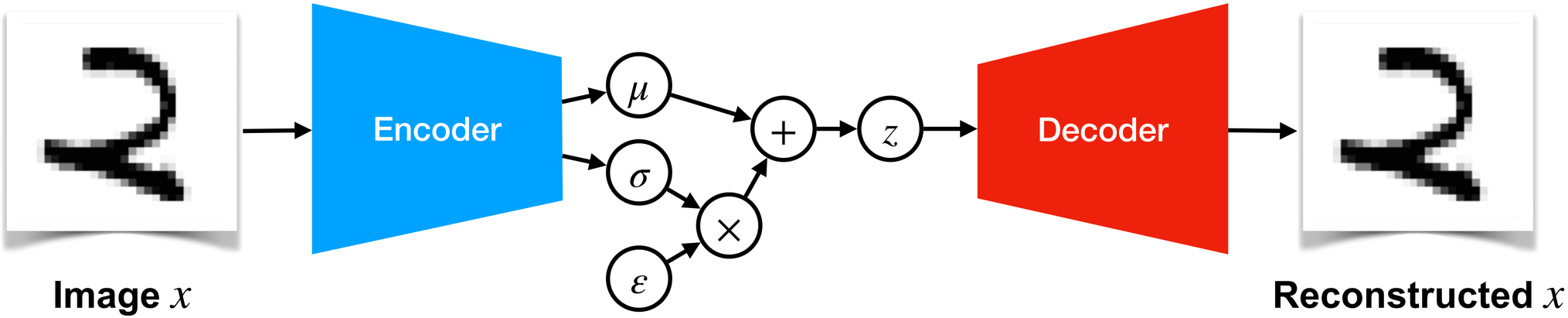
Hands-on with variational autoencoders



Variational autoencoders



$$\mathscr{L} = -\mathbb{E}_{x\sim p;z\sim p_\theta}\left[\log q_\phi(x\,|\,z)\right] + D_{KL}\left[p_\theta(z\,|\,x)\,\|\,q(z)\right]$$



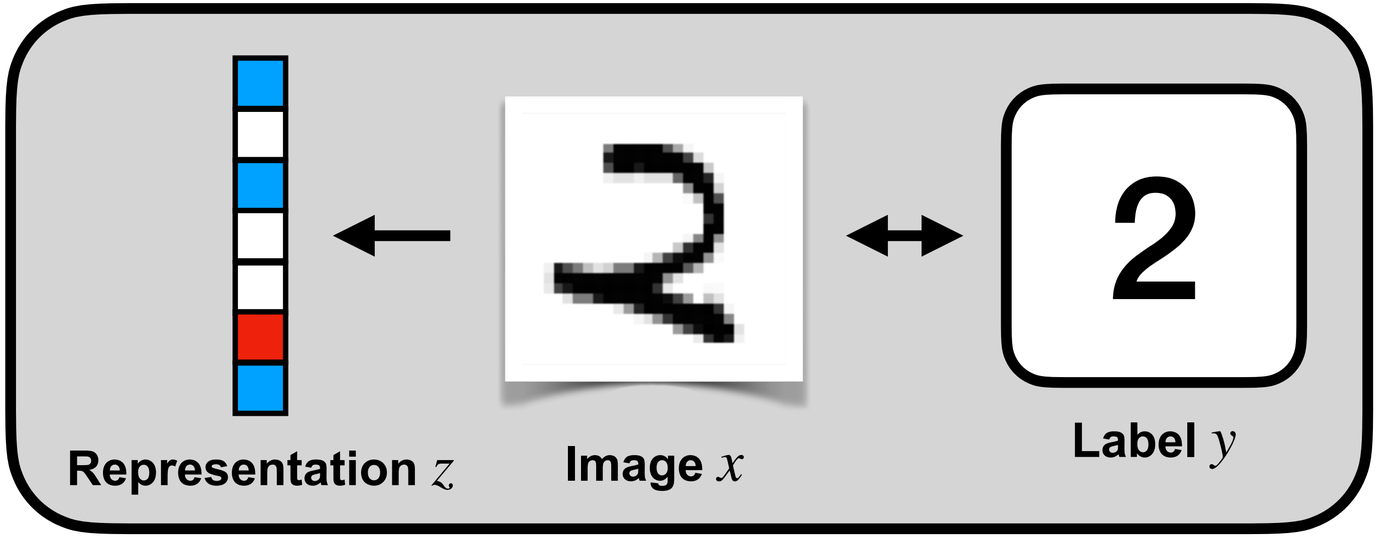
Semi-supervised learning

$$D_{KL}\big(P \,\|\, Q\big) = D_{KL}\big(p(x,y)\; p_\theta(z\,|\,x) \;\|\, q(z)\, q_\phi(y\,|\,z)\, q_\phi(x\,|\,z)\big)$$

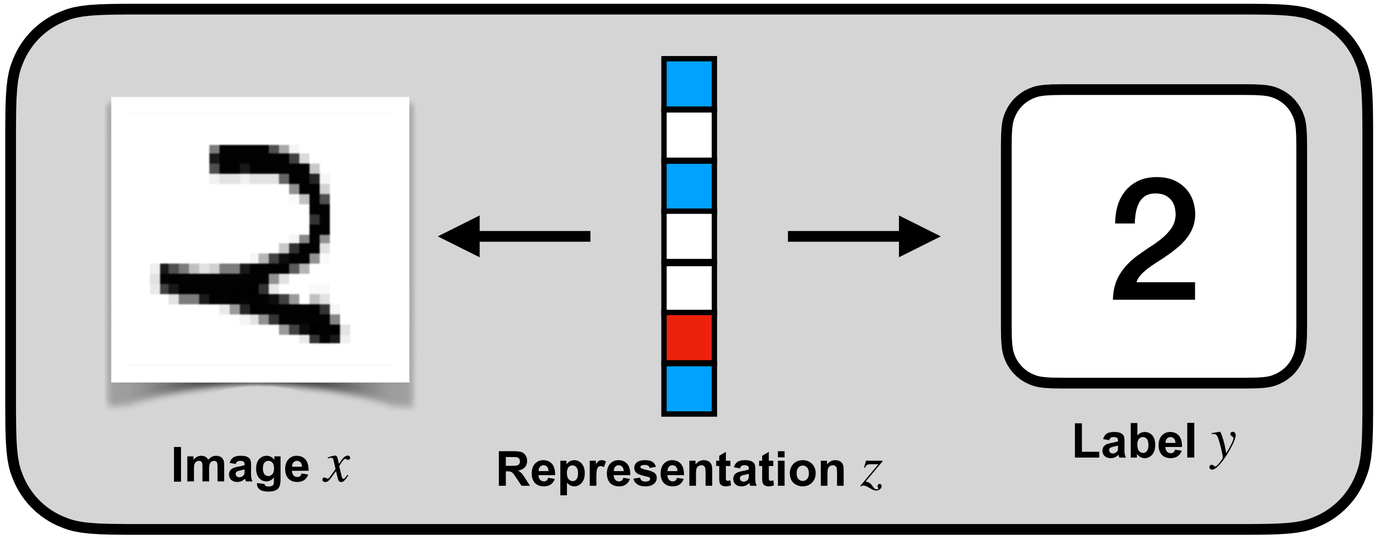
$$= \mathbb{E}_{x,y\sim p;z\sim p_\theta}\left[\log \frac{p(x,y)\, p_\theta(z\,|\,x)}{q(z)\, q_\phi(y\,|\,z)\, q_\phi(x\,|\,z)}\right]$$

$$= \mathbb{E}_{x,y\sim p;z\sim p_\theta}\big[\log p(x,y)\big]$$
$$-\mathbb{E}_{x,y\sim p;z\sim p_\theta}\big[\log q_\phi(y\,|\,z)\big] \quad \leftarrow \text{supervised}$$
$$-\mathbb{E}_{x,y\sim p;z\sim p_\theta}\big[\log q_\phi(x\,|\,z)\big]$$
$$+\mathbb{E}_{x,y\sim p;z\sim p_\theta}\left[\log \frac{p_\theta(z\,|\,x)}{q(z)}\right] \quad \leftarrow \text{unsupervised}$$

Real world P : $p(x,y,z) = p(x,y)\, p_\theta(z\,|\,x)$



Fantasy world: Q : $q(x,y,z) = q(z)\, q_{\phi_1}(y\,|\,z)\, q_{\phi_2}(x\,|\,z)$



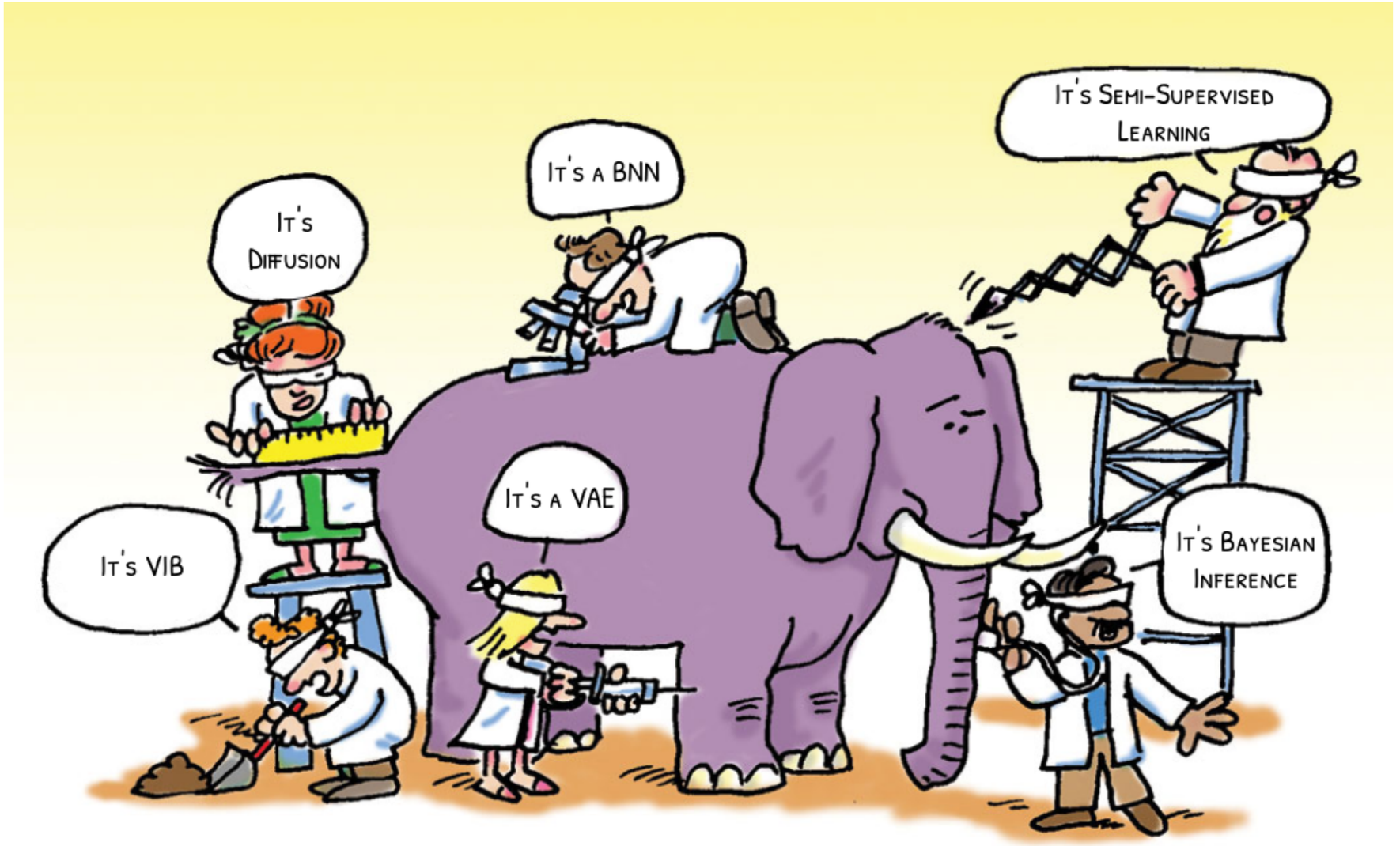


diagram shamelessly stolen from “KL is all you need,” A. Alemi

It’s $D_{KL}\left[P \,\|\, Q\right] = \mathbb{E}_P\left[\dfrac{\log P(x)}{\log Q(x)}\right] \geq 0$

But is KL Divergence all you need?

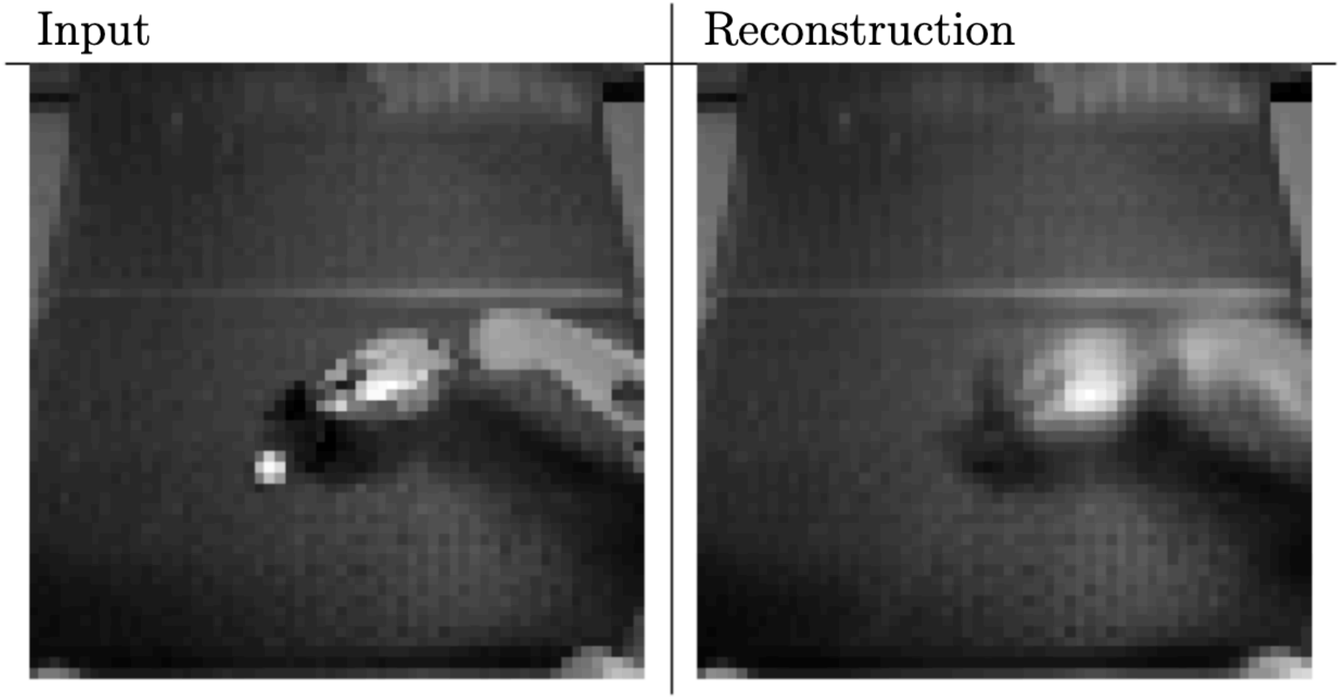


Figure 15.5: An autoencoder trained with mean squared error for a robotics task has failed to reconstruct a ping pong ball. The existence of the ping pong ball and all its spatial coordinates are important underlying causal factors that generate the image and are relevant to the robotics task. Unfortunately, the autoencoder has limited capacity, and the training with mean squared error did not identify the ping pong ball as being salient enough to encode. Images graciously provided by Chelsea Finn.

That’s all!

- Slides will be posted on the course website: https://colen-teaching.github.io/DATA442-Fall-2026
- Lab demos will also be posted
- ***PS4*** due November 25
- ***Final Presentations*** on December 8, 2-5pm, ISC 2280
- **REMEMBER:** TA Office Hours Tuesday 12-1pm, Student Success Center